AIM:

To Implement AND, OR and NAND Logic gates using a single layer perceptron

ALGORITHM:

* Activation function:

```
def Step_function(x):
    return np.where(x>=0,1,0)
```

* perception function

```
def perception(x, weights, bias):
    return. step_function(np.dot(x, weights) + bias)
```

Input Combination:

```
x = np.array([[0,0][0,1][1,0][1,1]])
```

1.) AND GATE:

```
weight_and = np.array([1,1])
bias_and = -1.5
output_and = perception(x, weights_and, bias_and)
print("AND Gate output:", output_and)
```

2.) OR GATE:

```
weight_or = np.array([1,1])
bias_or = -0.5
output_or = perceptron(x, weights_or, bias_or)
print("OR Gate output:", output_or)
```

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111) NAND Gate:

```
weight_nand = np.array([-1, -1])
bias_nand = 1.5
output_nand = perceptron(x, weights_nand, bias_nand)
print("NAND gate output", output_nand)
```

OUTPUT:

AND gate output: [0 0 0 1]

OR gate output: [0 1 1 1]

NAND gate output: [1 1 1 0]

Verification:

Input (x_1, x_2)	AND	OR	NAND
(0, 0)	0	0	1
(0, 1)	0	1	1
(1, 0)	0	1	1
(1, 1)	1	1	0

RESULT: